

Syntax Tree



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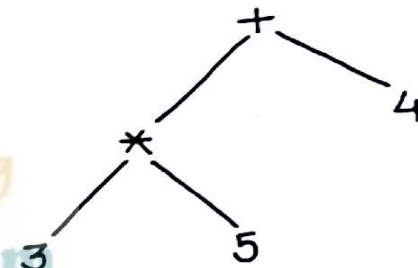
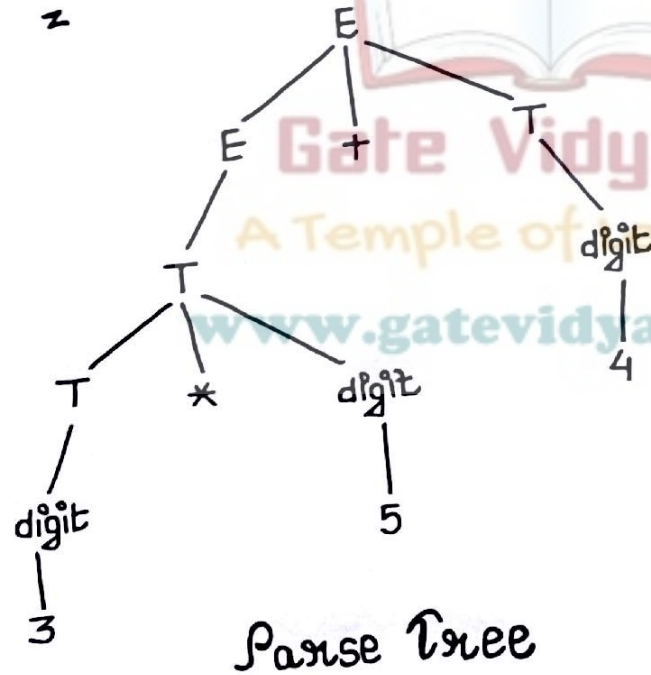
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Definition

A syntax tree (also known as abstract syntax tree) is a condensed form of parse tree.

Example:



Parse Tree Vs Syntax Tree

Parse Tree

A parse tree is a graphical representation of the replacement process in a derivation.

In parse trees,

- each internal node represents a grammar rule
- each leaf node represents a terminal

Syntax Tree

A syntax tree is a condensed form of parse tree.

In syntax trees,

- each internal node represents an operator
- each leaf node represents an operand

Parse trees represent every detail from the real syntax

Syntax trees does not represent every detail from the real syntax (that's why they are called abstract). For example- no rule nodes, no parentheses etc.

Parse trees are less dense compared to syntax trees for the same language construct

Syntax trees are more dense compared to parse trees for the same language construct.

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Problem-01:

Consider the following grammar-

$$E \rightarrow E + T / T$$

$$T \rightarrow T * F / F$$

$$F \rightarrow (E) / id$$

For the string,

$$w = id + id * id$$

Generate-

i> Parse tree

ii> Syntax tree

iii> DAG

Solution:-

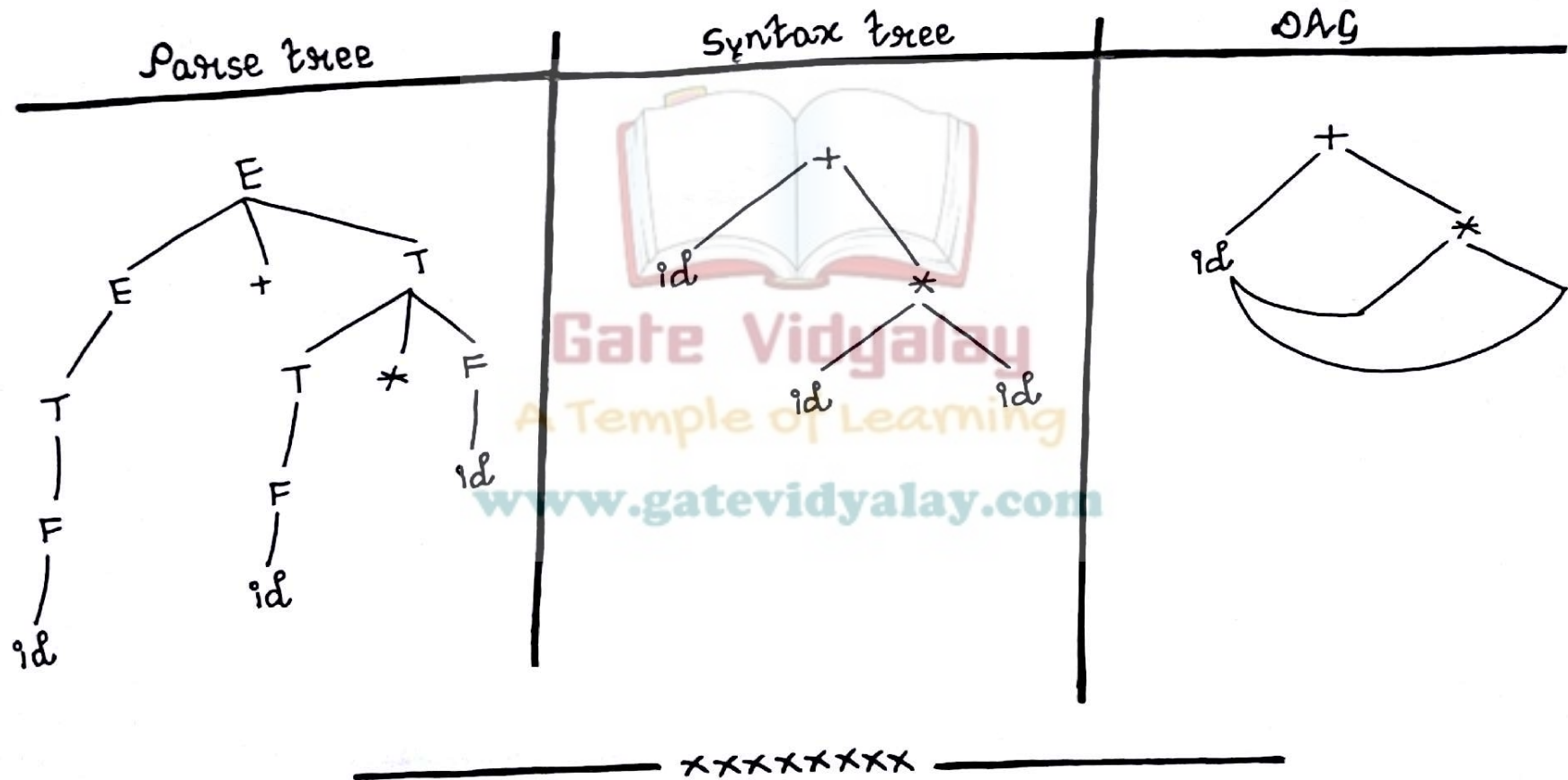
The given grammar is-

$$E \rightarrow E + T / T$$

$$T \rightarrow T * F / F$$

$$F \rightarrow (E) / id$$

$$w = id + id * id$$



Problem-02:-

Construct the syntax tree for the following expression -
 $(a+b) * (c-d) + ((e/f) * (a+b))$

Solution:-

Step-1:- Convert the given expression into a postfix expression:-

$$(a+b) * (c-d) + ((e/f) * (a+b))$$

$$\rightarrow ab+ * (c-d) + ((e/f) * (a+b))$$

$$\rightarrow ab+ * cd- + ((e/f) * (a+b))$$

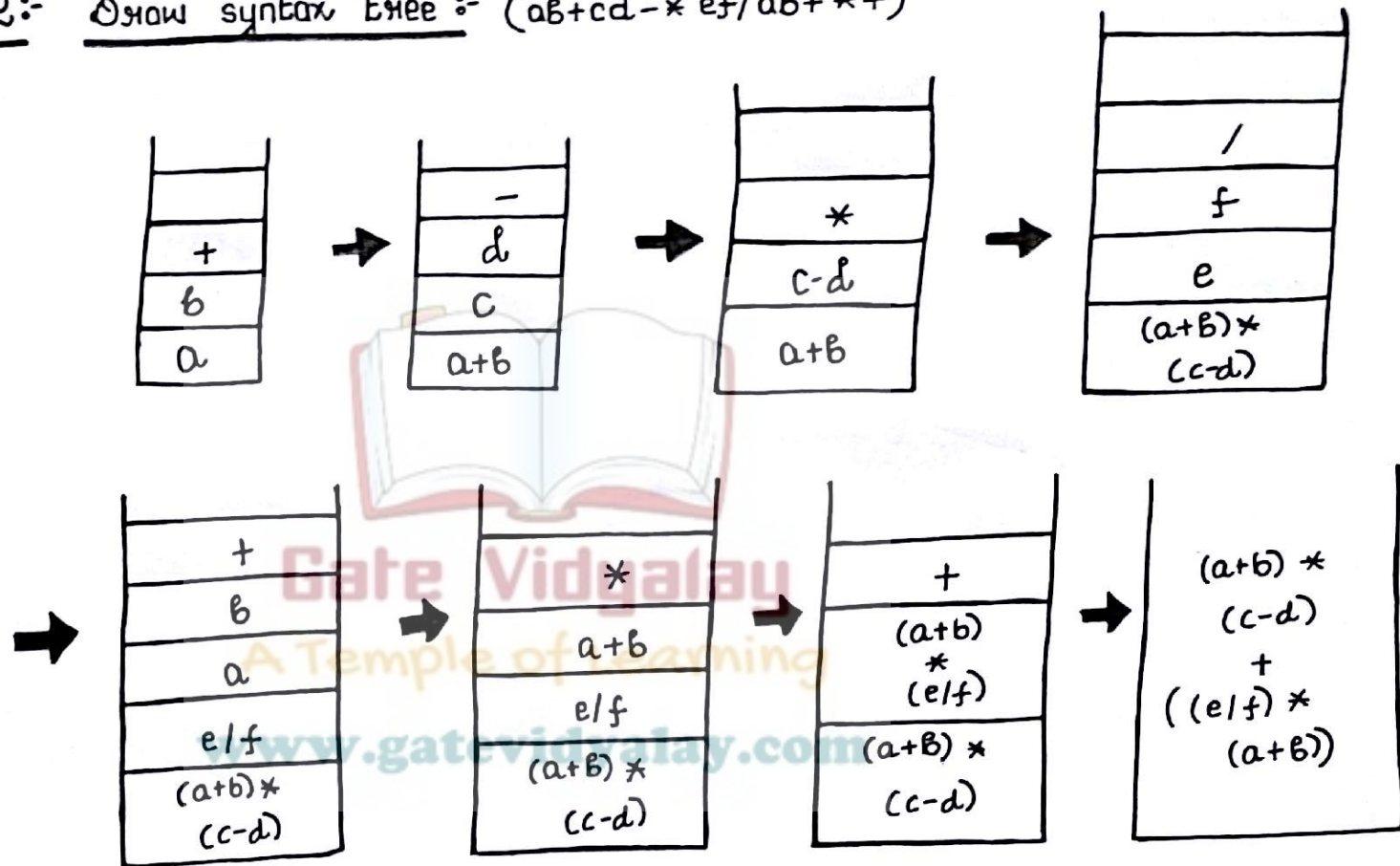
$$\rightarrow ab+ * cd- + (ef/ * ab+)$$

$$\rightarrow ab+ * cd- + ef/ab+*$$

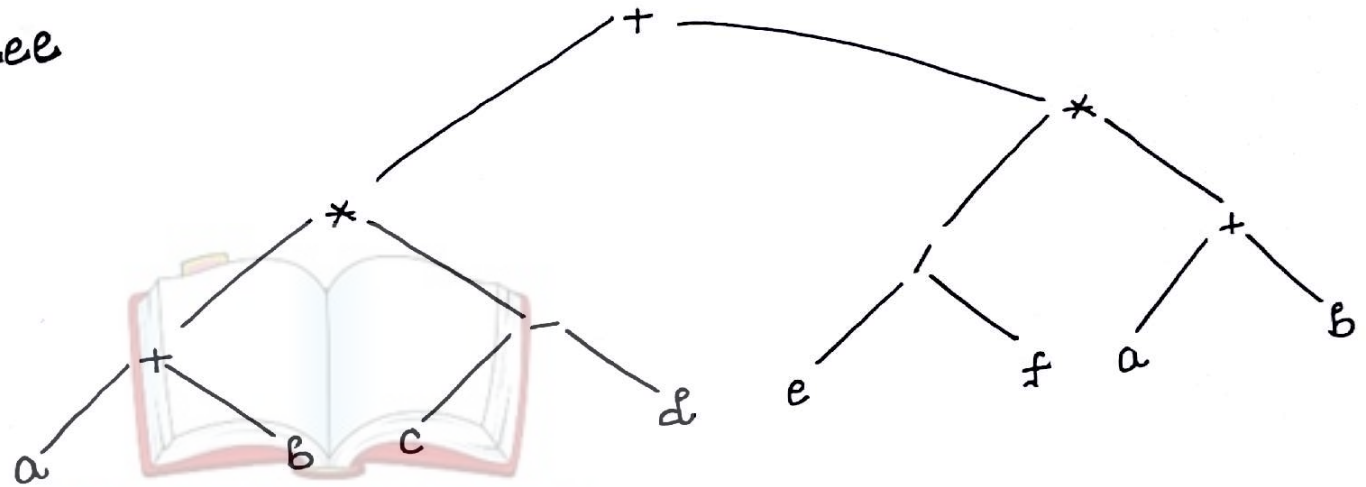
$$\rightarrow ab+cd-* + ef/ab+*$$

$$\rightarrow ab+cd-*ef/ab+*+$$

Step-2:- Draw syntax tree :- $(ab+cd-*ef/ab+*+)$



Syntax tree



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